RAG Project Assignment

E-Commerce Shopping Assistant (ShopKart)

Level: Beginner to Intermediate | Practical GenAI Project

1. Project Scenario

Build an AI-powered **E-Commerce RAG Assistant** for a fictional online shopping company called **ShopKart**.

Customers can ask questions about products, prices, delivery, returns, refunds, payment methods, and company policies. The chatbot must retrieve information from your own product catalog and company documents **before** generating answers.

2. Create Your Knowledge Base

Create these four `.txt` files. Do not use the previous company policy document.

File	Must contain
products.txt	Product name and ID, description, price, brand, category, availability, specifications
delivery_policy.txt	Delivery charges, standard delivery time, express delivery, tracking orders, delivery locations, delayed delivery policy
return_refund_policy.txt	Return eligibility, return window, refund processing time, replacement policy, damaged product procedure, non-returnable products
payment_policy.txt	UPI, credit and debit cards, cash on delivery, EMI, payment failures, refund payment methods

3. Sample Product Knowledge Base

Create `products.txt` with sample data like this:

```
SHOPKART PRODUCT CATALOG

PRODUCT 1
Product ID: SK101
Name: Samsung Galaxy M35
Category: Smartphone
Price: 19,999 INR
Brand: Samsung
RAM: 8GB
Storage: 128GB
Battery: 6000mAh
Display: Super AMOLED
Availability: In Stock
Warranty: 1 Year

PRODUCT 2
Product ID: SK102
Name: Redmi Note 14
Category: Smartphone
Price: 17,999 INR
Brand: Xiaomi
RAM: 8GB
Storage: 128GB
```

```
Battery: 5500mAh
Display: AMOLED
Availability: In Stock
Warranty: 1 Year

PRODUCT 3
Product ID: SK103
Name: HP Pavilion Laptop
Category: Laptop
Price: 59,999 INR
Brand: HP
RAM: 16GB
Storage: 512GB SSD
Processor: Intel Core i5
Availability: In Stock
Warranty: 1 Year

PRODUCT 4
Product ID: SK104
Name: boAt Airdopes 141
Category: Earbuds
Price: 1,299 INR
Brand: boAt
Battery Life: 42 Hours
Connectivity: Bluetooth
Availability: Out of Stock
Warranty: 1 Year
```

Note: These are fictional test records; do not treat them as current retail prices or verified specifications.

4. Technical Requirements

Use the same RAG technology stack you have been learning.

Technology	Purpose
Python	Backend
Sentence Transformers	Embeddings
all-MiniLM-L6-v2	Embedding model
NumPy	Vector similarity
Cosine Similarity	Semantic retrieval
Groq / OpenAI API	Answer generation (use whichever key you have)
FAISS (Bonus)	Vector database

5. Development Assignments

Task 1: Load multiple documents

Load all four .txt files into Python. Create a function:

```
def load_documents():
    pass
```

Task 2: Chunking

```
def chunk(text, size=50, overlap=10):
    pass
```

Requirements:

- Split each document into chunks.
- Preserve document filenames as metadata.
- Maintain chunk overlap.

Task 3: Generate embeddings

```
embedder = SentenceTransformer("all-MiniLM-L6-v2")
```

Generate normalized embeddings for all chunks.

Task 4: Semantic retrieval

```
def retrieve(question, top_k=3):
    pass
```

Retrieve the most relevant product or policy information.

Task 5: Generate answers

```
def answer(question):
    pass
```

Requirements:

- Retrieve relevant chunks.
- Construct context.
- Send context to the LLM.
- Return a grounded answer.
- Show source document names.

Task 6: Handle unknown questions

If a customer asks about a product that is not in the knowledge base, the assistant must say it cannot find that information.

6. Test Questions for Your RAG Application

Product-related questions

#	Question
1	What is the price of Samsung Galaxy M35?
2	Which smartphones have 8GB RAM?
3	Compare Samsung Galaxy M35 and Redmi Note 14.
4	Which products are currently out of stock?
5	Which laptop has 16GB RAM?
6	What is the battery capacity of Samsung Galaxy M35?

Delivery-related questions

#	Question
7	How many days does standard delivery take?
8	Is express delivery available?
9	How can I track my order?
10	What happens if my delivery is delayed?

Return and refund questions

#	Question
11	What is the return window?
12	How long does a refund take?
13	Can I return a damaged product?
14	Which products cannot be returned?

Payment-related questions

#	Question
15	Does ShopKart support UPI?
16	Is cash on delivery available?
17	What happens if a payment fails?

Advanced reasoning questions

#	Question
18	Which smartphone is cheaper, Samsung or Redmi?
19	Find an in-stock product under Rs. 20,000.
20	Recommend a smartphone with 8GB RAM and explain why.

Hallucination testing

#	Question
21	What is the price of iPhone 17 Pro?
22	Does ShopKart sell refrigerators?
23	What is the CEO's phone number?

Expected behavior: The assistant should not invent information that is absent from the knowledge base.

7. RAG Architecture

Four TXT knowledge base files (products, delivery, returns, payments)
Document loading + chunking
Sentence Transformer embeddings
Vector storage + metadata
Customer question
Semantic search + top-k retrieval
LLM + retrieved context
ShopKart AI Shopping Assistant

(Read top to bottom. Questions enter at step 5 and flow through steps 6 to 8.)

8. Bonus Challenges

After completing the basic project:

- Implement FAISS vector storage.
- Add metadata filtering by product category.
- Add product comparison.
- Implement similarity threshold.
- Display source citations.
- Build a Streamlit chat interface.
- Add conversation history.
- Support PDF product catalogs.

9. Evaluation

Criteria	Marks
Knowledge base creation	10
Multiple document loading	10
Chunking	10
Embedding generation	15
Semantic retrieval	20
LLM integration	15
Hallucination handling	10
Source citations	10
Total	**100**

Final goal: Build a working E-Commerce RAG chatbot that can answer product questions, compare products, retrieve company policies, and provide reliable answers using your own knowledge base.

Suggestion: First complete the project using NumPy similarity search. Then upgrade it to FAISS. This will help you understand why vector databases are useful in real-world RAG systems.